

INTEGRATED ELECTRICAL SYSTEMS OF SOLAR PHOTOVOLTAIC TECHNOLOGY

LABORATORY 5

GROUP 2



TÉCNICO LISBOA

Spectral Dependence of Photovoltaic Module Performance

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1 Introduction

The main objective of this laboratory experiment is to analyse the spectral dependence of photovoltaic module performance under different light wavelengths. Since the energy of incident photons depends on wavelength, the electrical response of a photovoltaic device is affected by the colour of the light source.

In this laboratory, the I–V and P–V characteristics of photovoltaic modules were obtained under red, green and blue light. In addition to the effect of colour, the experiment also considers different electrical configurations, partial shading conditions and the comparison between two monocrystalline photovoltaic modules.

The experimental results allow the evaluation of the influence of the incident spectrum on the main operating points of the photovoltaic devices, as well as the analysis of the effect of series connection, bypass diode operation and shading losses. This makes it possible to relate the spectral composition of light with the electrical behaviour of photovoltaic modules under different operating conditions.

2 Theoretical Background

2.1 Photon Energy and the Photoelectric Effect

The energy carried by a photon is determined by its wavelength λ according to:

$$E = \frac{hc}{\lambda} \quad (1)$$

where $h = 6.626 \times 10^{-34}$ J s is Planck's constant and $c = 3 \times 10^8$ m s⁻¹ is the speed of light in vacuum. Shorter wavelengths carry more energy per photon, blue light ($\lambda \approx 450$ nm) carries approximately 2.76 eV per photon, green light ($\lambda \approx 530$ nm) approximately 2.34 eV, and red light ($\lambda \approx 650$ nm) approximately 1.91 eV.

For a photon to generate an electron–hole pair through the internal photoelectric effect, its energy must exceed the bandgap energy W_G of the semiconductor:

$$hf \geq W_G \quad (2)$$

For monocrystalline silicon, $W_G \approx 1.12$ eV at room temperature, meaning all three colours satisfy this condition. However, the fraction of photons that are absorbed and converted into carriers depends on the optical absorption properties of the material.

2.2 Optical Absorption and Spectral Dependence

The absorption of light in a semiconductor follows the Lambert–Beer law. The photon flux ϕ decays exponentially with depth x into the material:

$$\phi(x) = \phi_0 e^{-\alpha x} \quad (3)$$

where $\phi_0 = \phi(x=0)(1 - R)$ is the photon flux at the surface after accounting for interface reflectance R , and α is the wavelength-dependent absorption coefficient of the material. The irradiance decays accordingly:

$$I_{ph}(x) = I_{ph0} e^{-\alpha x} \quad (4)$$

The rate of electron–hole pair generation due to the internal photoelectric effect, G_{fe} , is given by the spatial derivative of the photon flux:

$$G_{fe} = -\frac{\partial \phi}{\partial x} = \alpha \phi_0 e^{-\alpha x} \quad (5)$$

This shows that the generation rate is directly proportional to α . As shown in the figure, the absorption coefficient of silicon, represented by the red line, is significantly higher for shorter wavelengths. Blue light has a significantly higher absorption coefficient and is therefore absorbed within a thin surface layer, while red light has a lower α and penetrates deeper into the material before being fully absorbed. Given that the active junction region of a PV cell has a finite thickness, a fraction of the red photons may pass through without generating electron-hole pairs, resulting in a lower overall photocurrent compared to blue light under the same irradiance conditions.

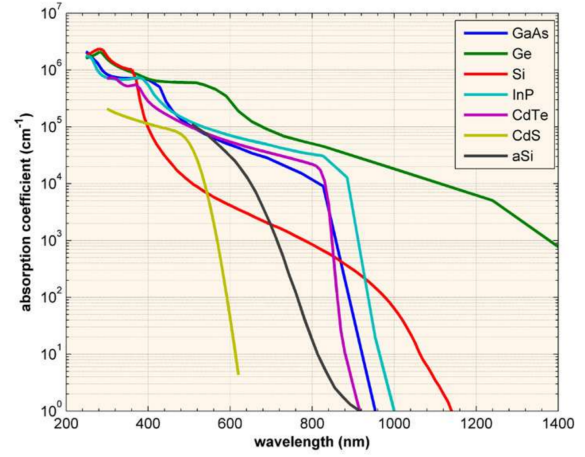


Figure 1: Absorption coefficient vs wavelength from theoretical slides.

The fraction of incident photons that effectively contribute to photogeneration is described by the quantum efficiency:

$$\eta_{qe} = (1 - R) (1 - e^{-\alpha W}) \quad (6)$$

where W is the thickness of the semiconductor layer. Near unity efficiency is achieved when $W > 4/\alpha$. Since α is larger for blue light, a thinner layer suffices to absorb it fully, whereas red light requires a proportionally thicker layer to achieve the same absorption fraction.

2.3 Impact on Key Operating Points

The one-diode equivalent circuit model (1D5P) describes the current-voltage relationship of a photovoltaic cell as:

$$I = I_{ph} - I_{is} \left(e^{\frac{V - R_s I}{n u_T}} - 1 \right) - \frac{V + R_s I}{R_{sh}} \quad (7)$$

where I_{ph} is the photogenerated current, I_{is} is the diode reverse saturation current, n is the ideality factor, $u_T = kT/q$ is the thermal voltage, R_s is the series resistance and R_{sh} is the shunt resistance.

The **short-circuit current** I_{sc} is obtained at $V = 0$, giving $I_{sc} \approx I_{ph}$. Since the photogenerated current is proportional to the carrier generation rate G_{fe} , which depends directly on α at the incident wavelength, light colours with a higher absorption coefficient are expected to produce larger I_{sc} values.

The **open-circuit voltage** V_{oc} is obtained at $I = 0$:

$$V_{oc} = n u_T \ln \left(\frac{I_{ph}}{I_{is}} + 1 \right) \quad (8)$$

Since V_{oc} depends logarithmically on I_{ph} , a higher photocurrent, produced by better-absorbed photons, also leads to a higher open-circuit voltage, though the dependence is weaker than for I_{sc} .

The **maximum power point** (MPP) is the operating point (V_M, I_M) at which $P = VI$ is maximised, found by imposing $dP/dV = 0$, giving $P_{max} = V_M I_M$.

The **fill factor** quantifies how close the I-V curve is to an ideal rectangle:

$$FF = \frac{P_{max}}{V_{oc} I_{sc}} = \frac{V_M I_M}{V_{oc} I_{sc}} \quad (9)$$

The **power conversion efficiency** is the ratio of maximum electrical power output to total incident optical power:

$$\eta = \frac{P_{max}}{G \cdot A} = \frac{V_M I_M}{G \cdot A} \quad (10)$$

where G is the irradiance and A is the active cell area.

These relations suggest that, under the same irradiance, blue light should yield the highest I_{sc} , V_{oc} and P_{max} , followed by green and then red light — a prediction that will be compared against the experimental results in Section 3.

3 Results

The experimental procedure followed the same load sweep methodology used in the previous laboratory classes to obtain the I–V and P–V characteristics of the photovoltaic modules. In this laboratory, the measurements were carried out under red, green and blue light, under constant irradiance conditions $G = 30 \text{ W m}^{-2}$ for different experimental configurations, namely a single monocrystalline module, two cells connected in series, the same series connection under 50% shading, the shaded series connection with bypass diode, and a second monocrystalline module.

From the obtained curves, the main operating points were determined and compared in order to analyse four main topics: the effect of different colours, the influence of module connection, the impact of shading, and the differences between the two tested photovoltaic modules.

3.1 0% Shading - (Monocrystalline - Seed Technology)

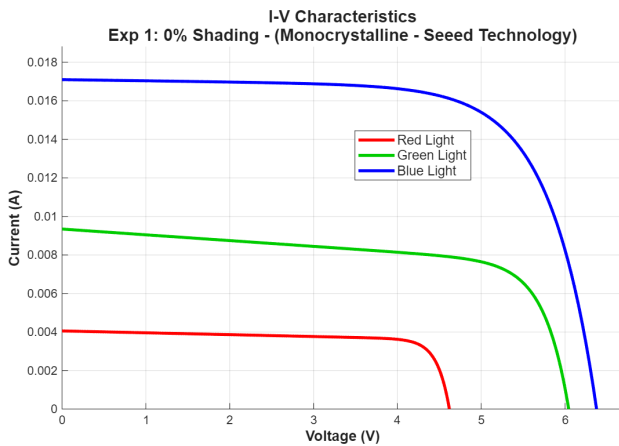


Figure 2: I–V characteristic under 0% shading (Seed Technology monocrystalline module)

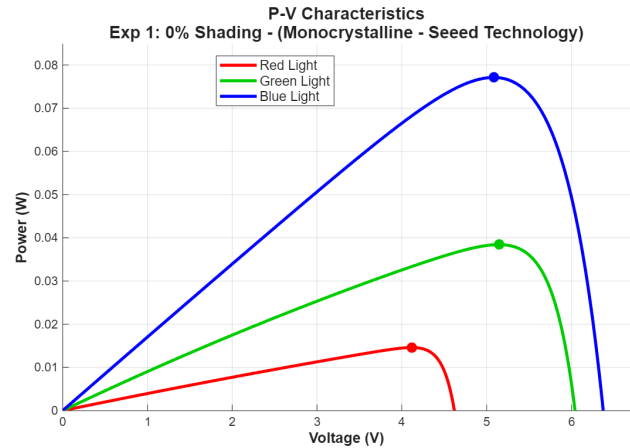


Figure 3: P–V characteristic under 0% shading (Seed Technology monocrystalline module)

3.2 0% Shading - 2 Cells in Series

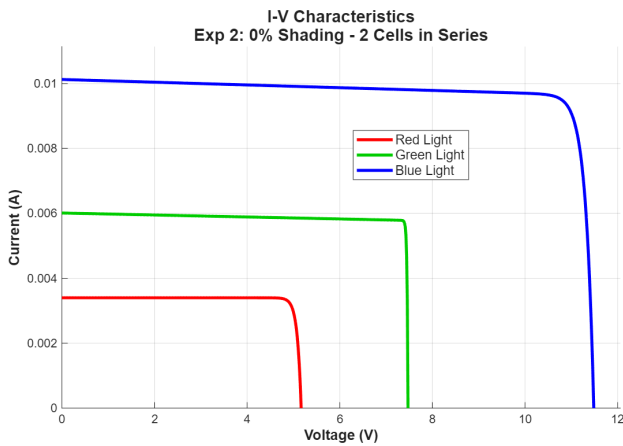


Figure 4: I–V characteristic of two cells connected in series under 0% shading

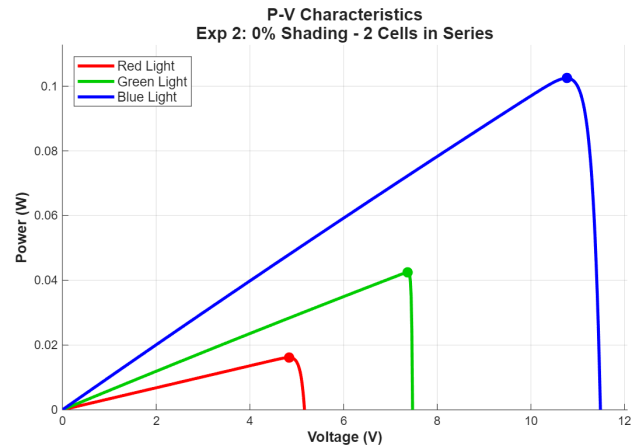


Figure 5: P–V characteristic of two cells connected in series under 0% shading

3.3 50% Shading - 2 Cells in Series

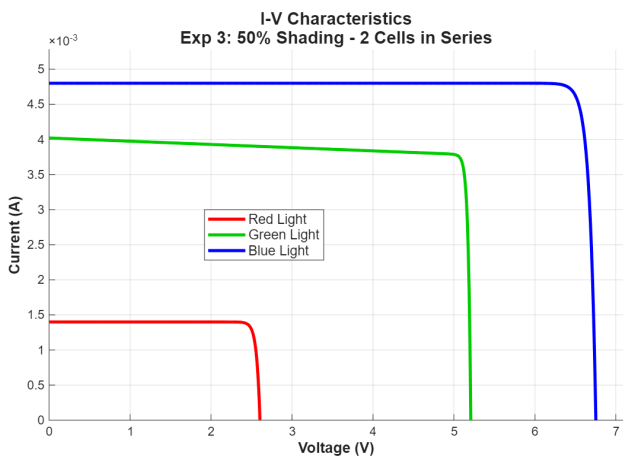


Figure 6: I–V characteristic of two cells in series under 50% shading

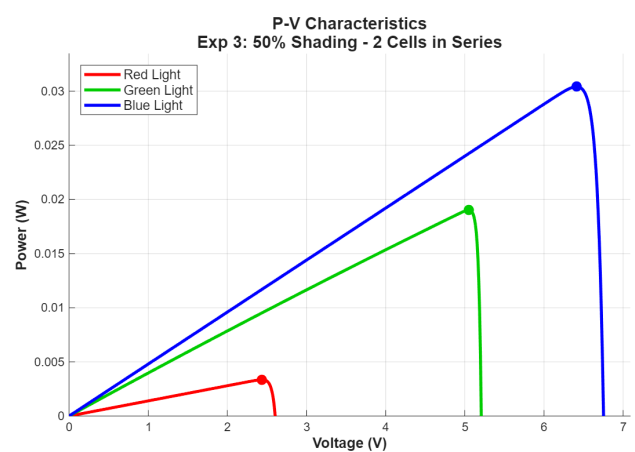


Figure 7: P–V characteristic of two cells in series under 50% shading

3.4 50% Shading - Series w/ Bypass Diode

Although the theoretical step-like behaviour expected in the I–V curve with a bypass diode under partial shading was not clearly observed in the experimental results, the beneficial effect of the bypass diode is still evident through the recovery of current and maximum power. The absence of a well-defined step may be related to the relatively low irradiance levels used in the experiment, the small current values involved, and the limited resolution of the measurement setup. In addition, the MATLAB curve fitting may have smoothed small discontinuities that would otherwise be more visible in the raw data.

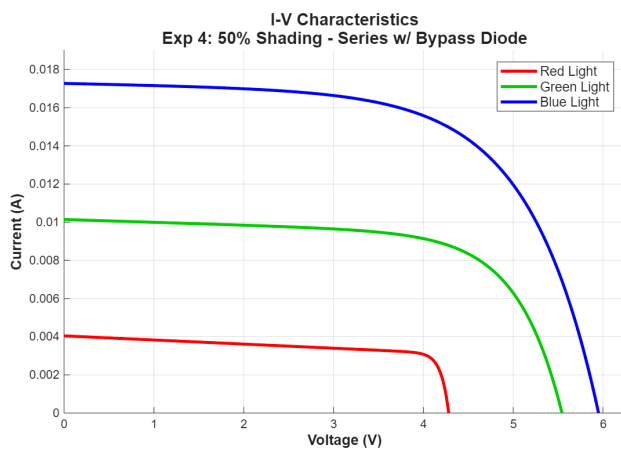


Figure 8: I–V characteristic of series-connected cells with bypass diode under 50% shading

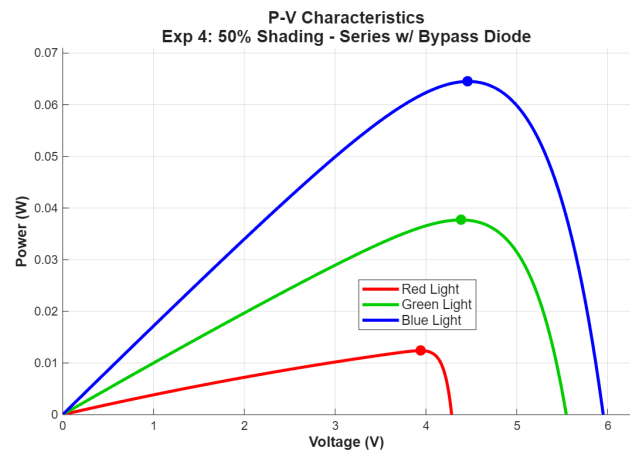


Figure 9: P–V characteristic of series-connected cells with bypass diode under 50% shading

3.5 0% Shading - (Monocrystalline - OSEPP)

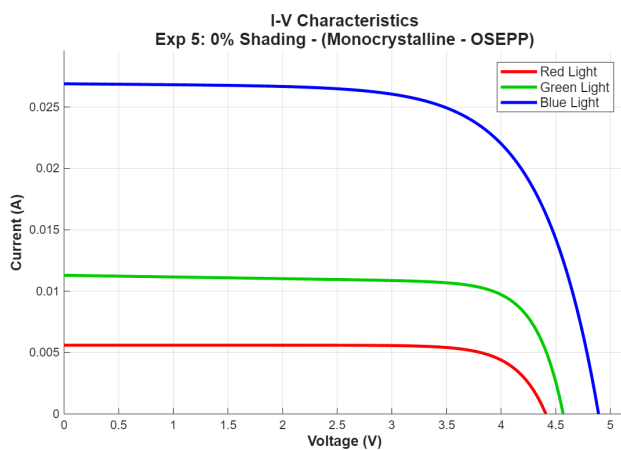


Figure 10: I–V characteristic under 0% shading (OSEPP monocrystalline module)

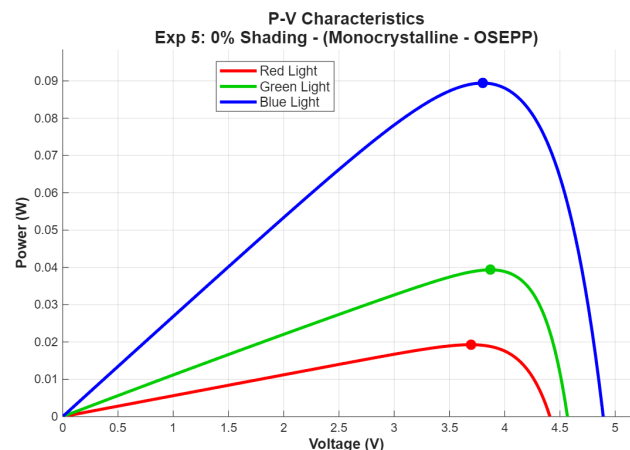


Figure 11: P–V characteristic under 0% shading (OSEPP monocrystalline module)

3.6 Comparison

Table 1: Comparison of electrical parameters under different light wavelengths and experimental setups

Experiment	Light	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	η (%)
Exp 1 (Seed, 0%)	Red	4.6200	0.0041	0.0146	4.1200	0.0035	77.95	12.64
	Green	6.0433	0.0093	0.0384	5.1471	0.0075	68.05	33.27
	Blue	6.3755	0.0171	0.0771	5.0851	0.0152	70.74	66.77
Exp 2 (Series, 0%)	Red	5.1660	0.0034	0.0162	4.8347	0.0033	91.98	6.99
	Green	7.4725	0.0060	0.0425	7.3676	0.0058	94.66	18.41
	Blue	11.4824	0.0101	0.1025	10.7691	0.0095	88.16	44.37
Exp 3 (Series, 50%)	Red	2.6040	0.0014	0.0034	2.4370	0.0014	92.06	1.45
	Green	5.2080	0.0040	0.0190	5.0514	0.0038	90.83	8.23
	Blue	6.7528	0.0048	0.0304	6.4145	0.0047	93.84	13.17
Exp 4 (Bypass, 50%)	Red	4.2823	0.0040	0.0124	3.9391	0.0031	71.53	5.36
	Green	5.5440	0.0101	0.0377	4.3885	0.0086	67.05	16.32
	Blue	5.9508	0.0173	0.0645	4.4601	0.0145	62.74	27.93
Exp 5 (OSEPP, 0%)	Red	4.4100	0.0056	0.0193	3.6941	0.0052	78.04	17.84
	Green	4.5684	0.0113	0.0393	3.8726	0.0102	76.25	36.43
	Blue	4.8922	0.0269	0.0894	3.8039	0.0235	67.95	82.78

Comparing two cells connected in series under 50% shading with a single cell under 50% shading, the series-connected configuration is expected to be more affected by the shading condition. In a single cell, the reduction in performance is mainly caused by the lower incident irradiance on that cell. However, in two cells connected in series, the shaded cell limits the current of the entire string, introducing mismatch losses and causing a stronger reduction in the overall output power. Therefore, although both configurations are negatively affected by 50% shading, the series-connected case presents a more severe degradation, since the effect of shading is no longer restricted to one cell but influences the behaviour of the whole circuit.

Comparing the two cells connected in series under 0% shading with the same configuration under 50% shading, it is possible to observe a significant reduction in electrical performance for all light colours. Under red light, V_{oc} decreases from 5.1660 V to 2.6040 V, I_{sc} from 0.0034 A to 0.0014 A, and P_{max} from 0.0162 W to 0.0034 W. Under green light, V_{oc} decreases from 7.4725 V to 5.2080 V, I_{sc} from 0.0060 A to 0.0040 A, and P_{max} from 0.0425 W to 0.0190 W. Under blue light, the same tendency is observed, with V_{oc} decreasing from 11.4824 V to 6.7528 V, I_{sc} from 0.0101 A to 0.0048 A, and P_{max} from 0.1025 W to 0.0304 W. These results confirm that partial shading strongly affects series-connected cells, since the shaded cell limits the current of the entire string, leading to mismatch losses and a significant reduction in the maximum output power. Although blue light still provides the highest power values in both cases, it is also under blue light that the absolute power reduction is more evident.

For 0% shading, two cells in series should present the same behaviour with or without bypass diodes, for red, green and blue light. This happens because, under uniform irradiance, the bypass diodes remain inactive and do not interfere with the normal operation of the cells.

Comparing two cells connected in series under 50% shading with the same configuration including a bypass diode, it is possible to observe a clear improvement in electrical performance for all light colours. Under red light, I_{sc} increases from 0.0014 A to 0.0040 A and P_{max} from 0.0034 W to 0.0124 W. Under green light, I_{sc} increases from 0.0040 A to 0.0101 A and P_{max} from 0.0190 W to 0.0377 W.

Under blue light, the same behaviour is observed, with I_{sc} increasing from 0.0048 A to 0.0173 A and P_{max} from 0.0304 W to 0.0645 W. These results confirm that the bypass diode mitigates the losses caused by partial shading, since it provides an alternative current path when the shaded cell limits the operation of the series string. Although the open-circuit voltage does not increase significantly, the recovery in current and power is evident for red, green and blue light.

Comparing the two photovoltaic modules under 0% shading, it is important to note that both of them are monocrystalline. Therefore, the differences observed in the experimental results should not be justified by differences in semiconductor technology, but rather by their nominal characteristics and construction. According to the datasheets, the Sseed module is rated for a higher voltage, while the OSEPP module presents lower nominal voltage but similar current capability. This tendency is consistent with the experimental results: the Sseed module presents higher voltage values, whereas the OSEPP module presents higher current values, especially under blue light. For example, under blue light, the Sseed module reaches $V_{oc} = 6.3755$ V and $I_{sc} = 0.0171$ A, while the OSEPP module reaches $V_{oc} = 4.8922$ V and $I_{sc} = 0.0269$ A. In terms of maximum power, the OSEPP module also performs slightly better, with $P_{max} = 0.0894$ W compared to 0.0771 W for the Sseed module. Therefore, although both cells are monocrystalline, their different nominal specifications lead to different electrical behaviours under the same spectral conditions.

4 Conclusion

This laboratory experiment analysed the spectral dependence of photovoltaic module performance under red, green and blue light, considering different electrical configurations and shading conditions.

For all the tested cases, blue light produced the highest current and maximum power, green light gave intermediate results, and red light led to the lowest performance. This result is consistent with the theoretical background presented in Section 2, since all three colours satisfy the bandgap condition of monocrystalline silicon, but blue light has a higher absorption coefficient and is therefore absorbed more effectively near the surface of the semiconductor, leading to a higher photogenerated current.

The comparison between a single module and two cells connected in series showed that the series connection increases the output voltage and may also increase the maximum power. This is in agreement with the expected behaviour of series-connected photovoltaic cells, where the voltages of the individual cells are added.

The introduction of 50% shading in the series string caused a significant reduction in performance, confirming the strong impact of mismatch losses in series-connected cells. The use of a bypass diode under 50% shading improved the current and power output for all colours, showing its important role in mitigating shading losses by providing an alternative path for the current.

Finally, the comparison between the Sseed and OSEPP modules showed that, although both are monocrystalline, they present different electrical behaviours under the same spectral conditions due to their different nominal characteristics. Overall, the results confirm that photovoltaic performance depends not only on the wavelength of the incident light, but also on the electrical configuration of the system, the presence of shading, and the module characteristics.